

Zweitore

Formeln, etc

v1

Quellen

- [1] Führer, Arnold; Heidemann, Klaus; Nerreter, Wolfgang (2019): Grundgebiete der Elektrotechnik. Band 1: Stationäre Vorgänge: mit 426 Bildern, 71 durchgerechneten Beispielen, 65 Praxisbezügen und 145 Aufgaben mit Lösungen / Arnold Führer † (1940-2010), Klaus Heidemann, Wolfgang Nerreter. 10., neu bearbeitete Auflage. München: Hanser.
- [2] Meier, Uwe; Stübbe, Oliver (2022): “Lineare Zweitore.” In: Elektrotechnik zum Selbststudium. Wiesbaden: Springer Fachmedien Wiesbaden, p. 335–368.
- [3] “2.3: Scattering Parameters” (2020): Engineering LibreTexts. 2.3: Scattering Parameters. Available at: URL: [https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Electronics/Microwave_and_RF_Design_III_-_Networks_\(Steer\)/02%3A_Chapter_2/2.3%3A_Scattering_Parameters](https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Electronics/Microwave_and_RF_Design_III_-_Networks_(Steer)/02%3A_Chapter_2/2.3%3A_Scattering_Parameters)

Z X H A Parameter

Z-Parameter

$$\begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} Z_{11} = \frac{U_1}{I_1} \Big|_{I_2=0} & Z_{12} = \frac{U_1}{I_2} \Big|_{I_1=0} \\ Z_{21} = \frac{U_2}{I_1} \Big|_{I_2=0} & Z_{22} = \frac{U_2}{I_2} \Big|_{I_1=0} \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

H-Parameter (Hybrid)

$$\begin{bmatrix} U_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} H_{11} = \frac{U_1}{I_1} \Big|_{U_2=0} & H_{12} = \frac{U_1}{U_2} \Big|_{I_1=0} \\ H_{21} = \frac{I_2}{I_1} \Big|_{U_2=0} & H_{22} = \frac{I_2}{U_2} \Big|_{I_1=0} \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ U_2 \end{bmatrix}$$

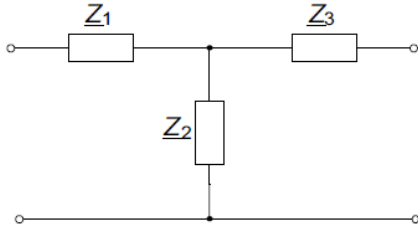
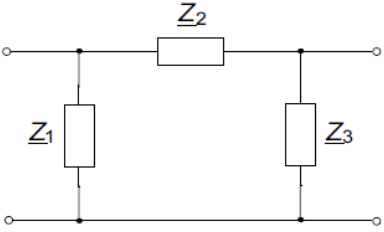
Y-Parameter

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} = \frac{I_1}{U_1} \Big|_{U_2=0} & Y_{12} = \frac{I_1}{U_2} \Big|_{U_1=0} \\ Y_{21} = \frac{I_2}{U_1} \Big|_{U_2=0} & Y_{22} = \frac{I_2}{U_2} \Big|_{U_1=0} \end{bmatrix} \cdot \begin{bmatrix} U_1 \\ U_2 \end{bmatrix}$$

Ketten-Parameter (engl. ABCD)

$$\begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_{11} = \frac{U_1}{U_2} \Big|_{I_2=0} & A_{12} = \frac{U_1}{-I_2} \Big|_{U_2=0} \\ A_{21} = \frac{I_1}{U_2} \Big|_{I_2=0} & A_{22} = \frac{I_1}{-I_2} \Big|_{U_2=0} \end{bmatrix} \cdot \begin{bmatrix} U_2 \\ -I_2 \end{bmatrix}$$

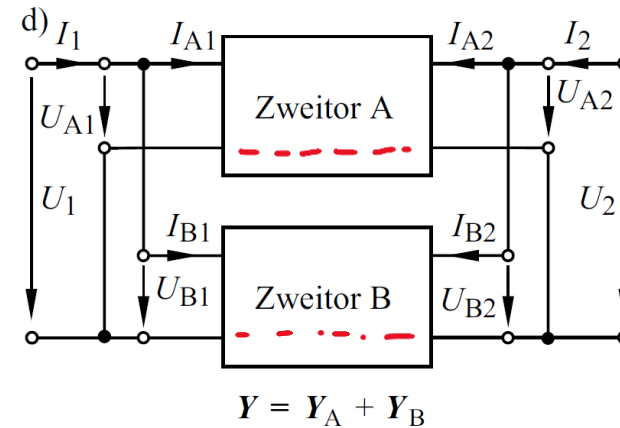
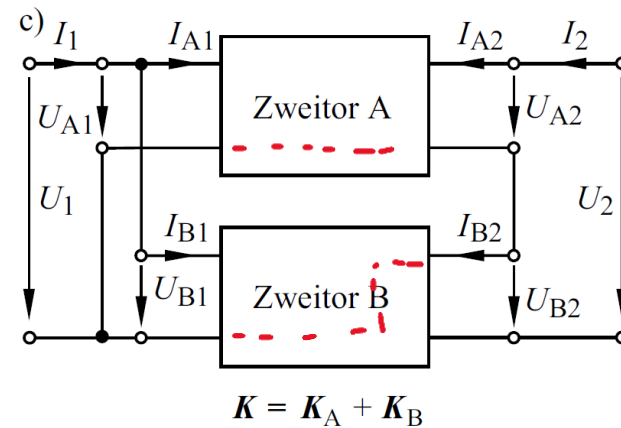
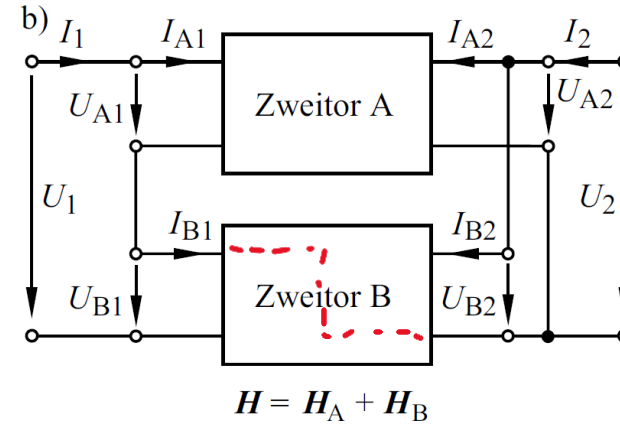
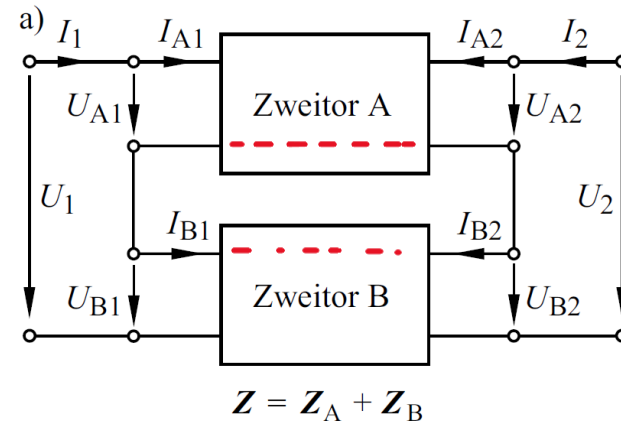
T-Schaltung – Pi Schaltung

	T-Vierpol	π -Vierpol
		
(Y)	$\underline{Y}_{11} = \frac{\underline{Z}_2 + \underline{Z}_3}{\underline{Z}_1 \underline{Z}_2 + \underline{Z}_1 \underline{Z}_3 + \underline{Z}_2 \underline{Z}_3}$ $\underline{Y}_{12} = \frac{-\underline{Z}_2}{\underline{Z}_1 \underline{Z}_2 + \underline{Z}_1 \underline{Z}_3 + \underline{Z}_2 \underline{Z}_3}$ $\underline{Y}_{21} = \frac{-\underline{Z}_2}{\underline{Z}_1 \underline{Z}_2 + \underline{Z}_1 \underline{Z}_3 + \underline{Z}_2 \underline{Z}_3}$ $\underline{Y}_{22} = \frac{\underline{Z}_1 + \underline{Z}_2}{\underline{Z}_1 \underline{Z}_2 + \underline{Z}_1 \underline{Z}_3 + \underline{Z}_2 \underline{Z}_3}$	$\underline{Y}_{11} = \frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2}$ $\underline{Y}_{12} = -\frac{1}{\underline{Z}_2}$ $\underline{Y}_{21} = -\frac{1}{\underline{Z}_2}$ $\underline{Y}_{22} = \frac{1}{\underline{Z}_2} + \frac{1}{\underline{Z}_3}$
(Z)	$\underline{Z}_{11} = \underline{Z}_1 + \underline{Z}_2$ $\underline{Z}_{12} = \underline{Z}_2$ $\underline{Z}_{21} = \underline{Z}_2$ $\underline{Z}_{22} = \underline{Z}_2 + \underline{Z}_3$	$\underline{Z}_{11} = \frac{\underline{Z}_1 (\underline{Z}_2 + \underline{Z}_3)}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3}$ $\underline{Z}_{12} = \frac{\underline{Z}_1 \underline{Z}_3}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3}$ $\underline{Z}_{21} = \frac{\underline{Z}_1 \underline{Z}_3}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3}$ $\underline{Z}_{22} = \frac{\underline{Z}_3 (\underline{Z}_1 + \underline{Z}_2)}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3}$

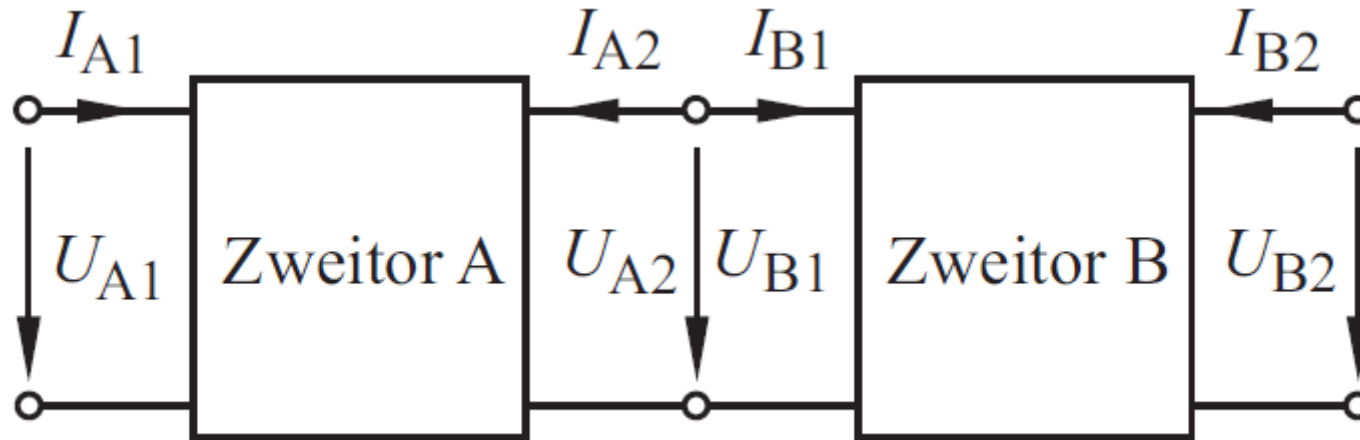
Umrechnungen

	[Z]	[Y]	[A]	[H]	[K]
[Z]	Z_{11} Z_{12} Z_{21} Z_{22}	$\frac{Y_{22}}{\det Y}$ $\frac{-Y_{12}}{\det Y}$ $\frac{-Y_{21}}{\det Y}$ $\frac{Y_{11}}{\det Y}$	$\frac{A_{11}}{A_{21}}$ $\frac{\det A}{A_{21}}$ $\frac{1}{A_{21}}$ $\frac{A_{22}}{A_{21}}$	$\frac{\det H}{H_{22}}$ $\frac{H_{12}}{H_{22}}$ $\frac{-H_{21}}{H_{22}}$ $\frac{1}{H_{22}}$	$\frac{1}{K_{11}}$ $\frac{-K_{12}}{K_{11}}$ $\frac{K_{21}}{K_{11}}$ $\frac{\det K}{K_{11}}$
[Y]	$\frac{Z_{22}}{\det Z}$ $\frac{-Z_{12}}{\det Z}$ $\frac{-Z_{21}}{\det Z}$ $\frac{Z_{11}}{\det Z}$	Y_{11} Y_{12} Y_{21} Y_{22}	$\frac{A_{22}}{A_{12}}$ $\frac{-\det A}{A_{12}}$ $\frac{-1}{A_{12}}$ $\frac{A_{11}}{A_{12}}$	$\frac{1}{H_{11}}$ $\frac{-H_{12}}{H_{11}}$ $\frac{H_{21}}{H_{11}}$ $\frac{\det H}{H_{11}}$	$\frac{\det K}{K_{22}}$ $\frac{K_{12}}{K_{22}}$ $\frac{-K_{21}}{K_{22}}$ $\frac{1}{K_{22}}$
[A]	$\frac{Z_{11}}{Z_{21}}$ $\frac{\det Z}{Z_{21}}$ $\frac{1}{Z_{21}}$ $\frac{Z_{22}}{Z_{21}}$	$\frac{-Y_{22}}{Y_{21}}$ $\frac{-1}{Y_{21}}$ $\frac{-\det Y}{Y_{21}}$ $\frac{-Y_{11}}{Y_{21}}$	A_{11} A_{12} A_{21} A_{22}	$\frac{-\det H}{H_{21}}$ $\frac{-H_{11}}{H_{21}}$ $\frac{-H_{22}}{H_{21}}$ $\frac{-1}{H_{21}}$	$\frac{1}{K_{21}}$ $\frac{K_{22}}{K_{21}}$ $\frac{K_{11}}{K_{21}}$ $\frac{\det K}{K_{21}}$
[H]	$\frac{\det Z}{Z_{22}}$ $\frac{Z_{12}}{Z_{22}}$ $\frac{-Z_{21}}{Z_{22}}$ $\frac{1}{Z_{22}}$	$\frac{1}{Y_{11}}$ $\frac{-Y_{12}}{Y_{11}}$ $\frac{Y_{21}}{Y_{11}}$ $\frac{\det Y}{Y_{11}}$	$\frac{A_{12}}{A_{22}}$ $\frac{\det A}{A_{22}}$ $\frac{-1}{A_{22}}$ $\frac{A_{21}}{A_{22}}$	H_{11} H_{12} H_{21} H_{22}	$\frac{K_{22}}{\det K}$ $\frac{-K_{12}}{\det K}$ $\frac{-K_{21}}{\det K}$ $\frac{K_{11}}{\det K}$
[K]	$\frac{1}{Z_{11}}$ $\frac{-Z_{12}}{Z_{11}}$ $\frac{Z_{21}}{Z_{11}}$ $\frac{\det Z}{Z_{11}}$	$\frac{\det Y}{Y_{22}}$ $\frac{Y_{12}}{Y_{22}}$ $\frac{-Y_{21}}{Y_{22}}$ $\frac{1}{Y_{22}}$	$\frac{A_{21}}{A_{11}}$ $\frac{-\det A}{A_{11}}$ $\frac{1}{A_{11}}$ $\frac{A_{12}}{A_{11}}$	$\frac{H_{22}}{\det H}$ $\frac{-H_{12}}{\det H}$ $\frac{-H_{21}}{\det H}$ $\frac{H_{11}}{\det H}$	K_{11} K_{12} K_{21} K_{22}

Ersatzschaltungen



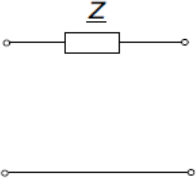
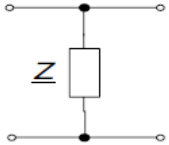
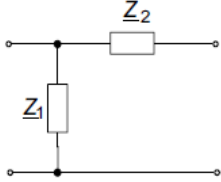
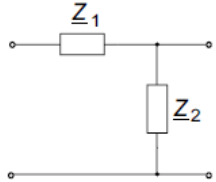
Kettenschaltung



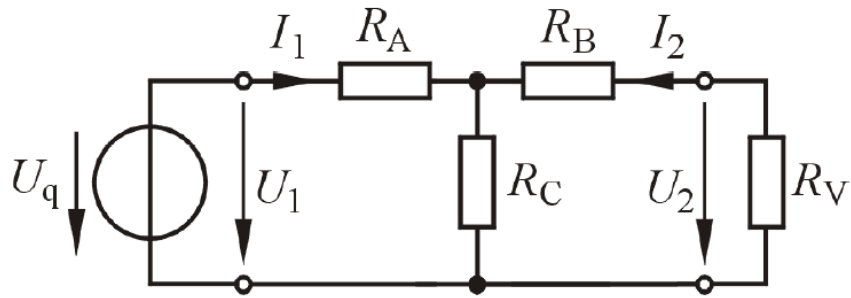
$$A = A_A \cdot A_B$$

Torbedingung ist automatisch erfüllt

Elementarzweipole

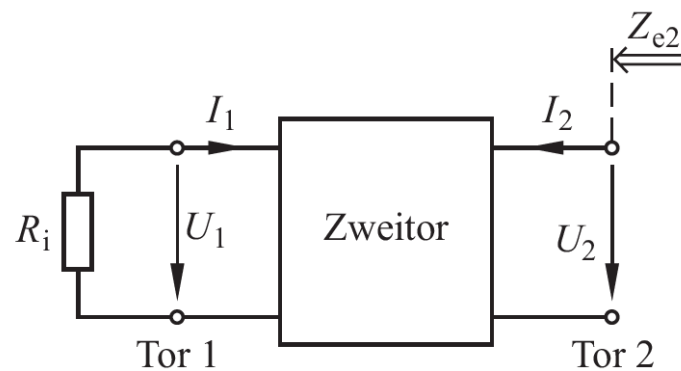
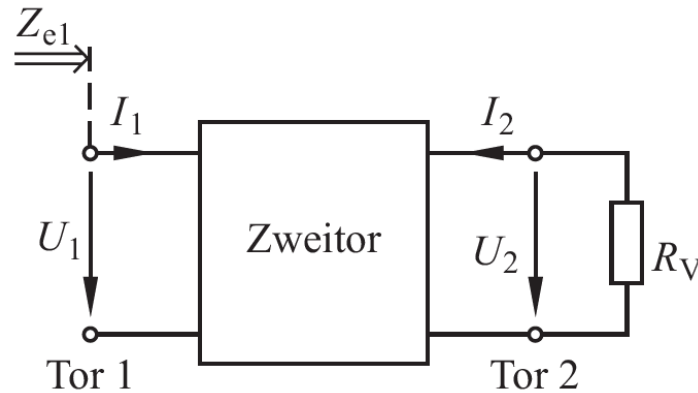
	Längswiderstand	Querwiderstand	Γ -Vierpol 1	Γ -Vierpol 2
				
(<u>Y</u>)	$\underline{Y}_{11} = \frac{1}{\underline{Z}}$ $\underline{Y}_{12} = -\frac{1}{\underline{Z}}$ $\underline{Y}_{21} = -\frac{1}{\underline{Z}}$ $\underline{Y}_{22} = \frac{1}{\underline{Z}}$	existieren nicht	$\underline{Y}_{11} = \frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2}$ $\underline{Y}_{12} = -\frac{1}{\underline{Z}_2}$ $\underline{Y}_{21} = -\frac{1}{\underline{Z}_2}$ $\underline{Y}_{22} = \frac{1}{\underline{Z}_2}$	$\underline{Y}_{11} = \frac{1}{\underline{Z}_1}$ $\underline{Y}_{12} = -\frac{1}{\underline{Z}_1}$ $\underline{Y}_{21} = -\frac{1}{\underline{Z}_1}$ $\underline{Y}_{22} = \frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2}$
(<u>Z</u>)	existieren nicht	$\underline{Z}_{11} = \underline{Z}$ $\underline{Z}_{12} = \underline{Z}$ $\underline{Z}_{21} = \underline{Z}$ $\underline{Z}_{22} = \underline{Z}$	$\underline{Z}_{11} = \underline{Z}_1$ $\underline{Z}_{12} = \underline{Z}_1$ $\underline{Z}_{21} = \underline{Z}_1$ $\underline{Z}_{22} = \underline{Z}_1 + \underline{Z}_2$	$\underline{Z}_{11} = \underline{Z}_1 + \underline{Z}_2$ $\underline{Z}_{12} = \underline{Z}_2$ $\underline{Z}_{21} = \underline{Z}_2$ $\underline{Z}_{22} = \underline{Z}_2$
(<u>A</u>)	$\underline{A}_{11} = 1$ $\underline{A}_{12} = \underline{Z}$ $\underline{A}_{21} = 0$ $\underline{A}_{22} = 1$	$\underline{A}_{11} = 1$ $\underline{A}_{12} = 0$ $\underline{A}_{21} = \frac{1}{\underline{Z}}$ $\underline{A}_{22} = 1$	$\underline{A}_{11} = 1$ $\underline{A}_{12} = \underline{Z}_2$ $\underline{A}_{21} = \frac{1}{\underline{Z}_1}$ $\underline{A}_{22} = 1 + \frac{\underline{Z}_2}{\underline{Z}_1}$	$\underline{A}_{11} = 1 + \frac{\underline{Z}_1}{\underline{Z}_2}$ $\underline{A}_{12} = \underline{Z}_1$ $\underline{A}_{21} = \frac{1}{\underline{Z}_2}$ $\underline{A}_{22} = 1$

Übertragungsfaktoren



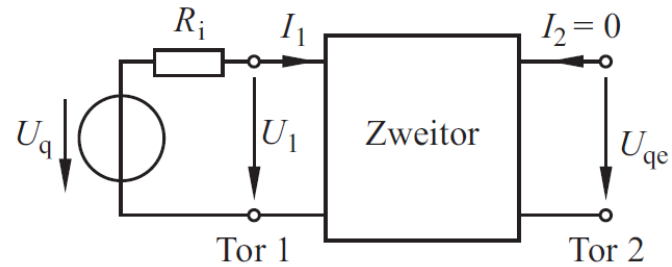
Z	$T_u = \frac{Z_{21}}{Z_{11} + G_V \cdot \det \mathbf{Z}}$	$T_i = \frac{-Z_{21}}{Z_{22} + R_V}$
Y	$T_u = \frac{-Y_{21}}{Y_{22} + G_V}$	$T_i = \frac{Y_{21}}{Y_{11} + R_V \cdot \det \mathbf{Y}}$
A	$T_u = \frac{1}{A_{11} + G_V A_{12}}$	$T_i = \frac{-1}{R_V A_{21} + A_{22}}$
H	$T_u = \frac{-H_{21}}{\det \mathbf{H} + G_V H_{11}}$	$T_i = \frac{H_{21}}{1 + R_V H_{22}}$
K	$T_u = \frac{K_{21}}{1 + G_V K_{22}}$	$T_i = \frac{-K_{21}}{\det \mathbf{K} + R_V K_{11}}$

Ersatzwiderstände



Z	$Z_{e1} = Z_{11} - \frac{Z_{12}Z_{21}}{Z_{22} + R_V}$	$Z_{e2} = Z_{22} - \frac{Z_{12}Z_{21}}{Z_{11} + R_i}$
Y	$Y_{e1} = Y_{11} - \frac{Y_{12}Y_{21}}{Y_{22} + G_V}$	$Y_{e2} = Y_{22} - \frac{Y_{12}Y_{21}}{Y_{11} + G_i}$
A	$Z_{e1} = \frac{A_{11}R_V + A_{12}}{A_{21}R_V + A_{22}}$	$Z_{e2} = \frac{A_{22}R_i + A_{12}}{A_{21}R_i + A_{11}}$
H	$Z_{e1} = H_{11} - \frac{H_{12}H_{21}}{H_{22} + G_V}$	$Y_{e2} = H_{22} - \frac{H_{12}H_{21}}{H_{11} + R_i}$
K	$Y_{e1} = K_{11} - \frac{K_{12}K_{21}}{K_{22} + R_V}$	$Z_{e2} = K_{22} - \frac{K_{12}K_{21}}{K_{11} + G_i}$

Ersatzquelle



Z	$U_{qe} = \frac{Z_{21} U_q}{Z_{11} + R_i}$	$I_{qe} = \frac{-Y_{21} I_q}{Y_{11} + G_i}$	Y
A	$U_{qe} = \frac{U_q}{R_i \cdot A_{21} + A_{11}}$		
H	$I_{qe} = \frac{-H_{21} U_q}{H_{11} + R_i}$	$U_{qe} = \frac{K_{21} I_q}{K_{11} + G_i}$	K

Symmetriebedingungen

Matrix	Längssymmetrie	Übertragungssymmetrie
$[Z]$	$\underline{Z}_{11} = \underline{Z}_{22}, \underline{Z}_{12} = \underline{Z}_{21}$	$\underline{Z}_{12} = \underline{Z}_{21}$
$[Y]$	$\underline{Y}_{11} = \underline{Y}_{22}, \underline{Y}_{12} = \underline{Y}_{21}$	$\underline{Y}_{12} = \underline{Y}_{21}$
$[A]$	$\underline{A}_{11} = \underline{A}_{22}, \det [A] = 1$	$\det [A] = 1$
$[H]$	$\det [H] = 1, \underline{H}_{12} = -\underline{H}_{21}$	$\underline{H}_{12} = -\underline{H}_{21}$
$[K]$	$\det [K] = 1, \underline{K}_{12} = -\underline{K}_{21}$	$\underline{K}_{12} = -\underline{K}_{21}$

S-Parameter Umrechnung

	S	In terms of S
z	$Z'_{11} = z_{11} Z_0 \quad Z'_{12} = z_{12} Z_0$	$Z'_{21} = z_{21} Z_0 \quad Z'_{22} = z_{22} Z_0$
	$\delta_z = (1 + z_{11})(1 + z_{22}) - z_{12}z_{21}$ $S_{11} = [(z_{11} - 1)(z_{22} + 1) - z_{12}z_{21}]/\delta_z$ $S_{12} = 2z_{12}/\delta_z$ $S_{21} = 2z_{21}/\delta_z$ $S_{22} = [(z_{11} + 1)(z_{22} - 1) - z_{12}z_{21}]/\delta_z$	$\delta_S = (1 - S_{11})(1 - S_{22}) - S_{12}S_{21}$ $z_{11} = [(1 + S_{11})(1 - S_{22}) + S_{12}S_{21}]/\delta_S$ $z_{12} = 2S_{12}/\delta_S$ $z_{21} = 2S_{21}/\delta_S$ $z_{22} = [(1 - S_{11})(1 + S_{22}) + S_{12}S_{21}]/\delta_S$
y	$Y'_{11} = y_{11}/Z_0 \quad Y'_{12} = y_{12}/Z_0$	$Y'_{21} = y_{21}/Z_0 \quad Y'_{22} = y_{22}/Z_0$
	$\delta_y = (1 + y_{11})(1 + y_{22}) - y_{12}y_{21}$ $S_{11} = [(1 - y_{11})(1 + y_{22}) + y_{12}y_{21}]/\delta_y$ $S_{12} = -2y_{12}/\delta_y$ $S_{21} = -2y_{21}/\delta_y$ $S_{22} = [(1 + y_{11})(1 - y_{22}) + y_{12}y_{21}]/\delta_y$	$\delta_S = (1 + S_{11})(1 + S_{22}) - S_{12}S_{21}$ $y_{11} = [(1 - S_{11})(1 + S_{22}) + S_{12}S_{21}]/\delta_S$ $y_{12} = -2S_{12}/\delta_S$ $y_{21} = -2S_{21}/\delta_S$ $y_{22} = [(1 + S_{11})(1 - S_{22}) + S_{12}S_{21}]/\delta_S$
h	$H'_{11} = h_{11} Z_0 \quad H'_{12} = h_{12}$	$H'_{21} = h_{21} \quad H'_{22} = h_{22}/Z_0$
	$\delta_h = (1 + h_{11})(1 + h_{22}) - h_{12}h_{21}$ $S_{11} = [(h_{11} - 1)(h_{22} + 1) - h_{12}h_{21}]/\delta_h$ $S_{12} = 2h_{12}/\delta_h$ $S_{21} = -2h_{21}/\delta_h$ $S_{22} = [(1 + h_{11})(1 - h_{22}) + h_{12}h_{21}]/\delta_h$	$\delta_S = (1 - S_{11})(1 + S_{22}) + S_{12}S_{21}$ $h_{11} = [(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}]/\delta_S$ $h_{12} = 2S_{12}/\delta_S$ $h_{21} = -2S_{21}/\delta_S$ $h_{22} = [(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}]/\delta_S$

S-Parameter Umrechnung

TABLE D.1 S-Parameter Conversion Chart

	S	z	y	$ABCD$
S_{11}	S_{11}	$\frac{(z_{11} - Z_0)(z_{22} + Z_0) - z_{12}z_{21}}{(z_{11} + Z_0)(z_{22} + Z_0) - z_{12}z_{21}}$	$\frac{(Y_0 - y_{11})(Y_0 + y_{22}) + y_{12}y_{21}}{(Y_0 + y_{11})(Y_0 + y_{22}) - y_{12}y_{21}}$	$\frac{A + B/Z_0 - CZ_0 - D}{A + B/Z_0 + CZ_0 + D}$
S_{12}	S_{12}	$\frac{2z_{12}Z_0}{(z_{11} + Z_0)(z_{22} + Z_0) - z_{12}z_{21}}$	$\frac{-2y_{12}Y_0}{(Y_0 + y_{11})(Y_0 + y_{22}) - y_{12}y_{21}}$	$\frac{2(AD - BC)}{A + B/Z_0 + CZ_0 + D}$
S_{21}	S_{21}	$\frac{(z_{11} + Z_0)(z_{22} + Z_0) - z_{12}z_{21}}{2z_{21}Z_0}$	$\frac{(Y_0 + y_{11})(Y_0 + y_{22}) - y_{12}y_{21}}{-2y_{21}Y_0}$	$\frac{2}{A + B/Z_0 + CZ_0 + D}$
S_{22}	S_{22}	$\frac{(z_{11} + Z_0)(z_{22} + Z_0) - z_{12}z_{21}}{(z_{11} + Z_0)(z_{22} - Z_0) - z_{12}z_{21}}$	$\frac{(Y_0 + y_{11})(Y_0 + y_{22}) - y_{12}y_{21}}{(Y_0 + y_{11})(Y_0 - y_{22}) + y_{12}y_{21}}$	$\frac{A + B/Z_0 + CZ_0 + D}{-A + B/Z_0 - CZ_0 + D}$
z_{11}	$Z_0 \frac{(1 + S_{11})(1 - S_{22}) + S_{12}S_{21}}{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}$	z_{11}	$\frac{y_{22}}{y_{11}y_{22} - y_{12}y_{21}}$	$\frac{A}{C}$
z_{12}	$Z_0 \frac{2S_{12}}{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}$	z_{12}	$\frac{-y_{12}}{y_{11}y_{22} - y_{12}y_{21}}$	$\frac{AD - BC}{C}$
z_{21}	$Z_0 \frac{2S_{21}}{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}$	z_{21}	$\frac{-y_{21}}{y_{11}y_{22} - y_{12}y_{21}}$	$\frac{1}{C}$
z_{22}	$Z_0 \frac{(1 - S_{11})(1 + S_{22}) + S_{12}S_{21}}{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}$	z_{22}	$\frac{y_{11}}{y_{11}y_{22} - y_{12}y_{21}}$	$\frac{D}{C}$
y_{11}	$Y_0 \frac{(1 - S_{11})(1 + S_{22}) + S_{12}S_{21}}{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}$	$\frac{z_{22}}{z_{11}z_{22} - z_{12}z_{21}}$	y_{11}	$\frac{D}{B}$
y_{12}	$Y_0 \frac{-2S_{12}}{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}$	$\frac{-z_{12}}{z_{11}z_{22} - z_{12}z_{21}}$	y_{12}	$\frac{BC - AD}{B}$
y_{21}	$Y_0 \frac{-2S_{21}}{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}$	$\frac{-z_{21}}{z_{11}z_{22} - z_{12}z_{21}}$	y_{21}	$\frac{-1}{B}$
y_{22}	$Y_0 \frac{(1 + S_{11})(1 - S_{22}) + S_{12}S_{21}}{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}$	$\frac{z_{11}}{z_{11}z_{22} - z_{12}z_{21}}$	y_{22}	$\frac{A}{B}$
A	$\frac{(1 + S_{11})(1 - S_{22}) + S_{12}S_{21}}{2S_{21}}$	$\frac{z_{11}}{z_{21}}$	$\frac{-y_{22}}{y_{21}}$	A
B	$Z_0 \frac{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}{2S_{21}}$	$\frac{z_{11}z_{22} - z_{12}z_{21}}{z_{21}}$	$\frac{-1}{y_{21}}$	B
C	$\frac{1}{Z_0} \frac{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}{2S_{21}}$	$\frac{1}{z_{21}}$	$-\frac{y_{11}y_{22} - y_{12}y_{21}}{y_{21}}$	C
D	$\frac{(1 - S_{11})(1 + S_{22}) - S_{12}S_{21}}{2S_{21}}$	$\frac{z_{22}}{z_{21}}$	$\frac{-y_{11}}{y_{21}}$	D